

PROJECT TRICORDER Mk I

Field operations manual · Revision 0.6.45

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Jim Ames, Claude and Grok · N2NHU Labs · Newburgh, New York 12550

RENT-A-HAL Android Edition, implemented as Project Tricorder Mk I.

1. Warnings

NAVIGATION. Not certified by FAA, NASA, EASA, or ICAO. The artificial horizon, G-meter, GPS speed, barometric altitude proxy, satellite map, star map, SOL orbits, moon phase, GNSS clock, HF/VHF/UHF forecasts, and the walking NAV aid **MUST NEVER** be used to fly, land, or navigate any craft or vehicle. Line-of-sight NAV is a walking survey sketch. Eyeballs own fences, creeks, and traffic. This load is not DO-178C airborne software.

WATCH. Weapon and gunfire phrases are caption keywords and YAMNet labels. Not a weapons detector. Not a 911 substitute.

MEDICAL / MEMORY ALPHA. MEDICAL is an encyclopedia briefing, not a clinician. MEMORY ALPHA is Simple English Wikipedia, not the Star Trek wiki.

MARKS. Not authorized by or affiliated with The Wand Company, Paramount SkyDance, Desilu, Gulf and Western, CBS, Roddenberry Entertainment, or Wah Chang.

2. What this unit is

Mk I is an integration, not an app. The APK is the face. Four Termux servers on loopback (chat :8080, vision :8081, image :8188, tts :5000) plus on-device TFLite packs (YAMNet, optional BirdNET, flora/fauna, translator) are the instrument. Reference computer: Galaxy Z Fold 3, Snapdragon 888, 12 GB.

This revision documents every operator-visible change since Rev 0.6.37, especially walking NAV (0.6.38–0.6.45), and keeps the earlier face, dictation, watch, sky, and backend sections current.

3. Changes since Rev 0.6.37 — walking NAV

3.1 H button cycle

H now cycles three states: off → horizon → NAV (horizon stays) → off. Yellow/green lamp plus the tiny label reads NAV while the plot is up. Voice: TRICORDER NAV opens the same pane.

3.2 What NAV is

Line-of-sight walking only. No road router, no turn-by-turn, not for driving or flight. The unit computes initial bearing and great-circle distance, draws a north-up symbol sketch (YOU, green trail of the walk, DEST, CAMP, waypoints), and chimes if cross-track error exceeds 30 m. GPS quiet more than 8 s: dead-reckon along magnetic/IMU heading at a walk, labeled DR.

3.3 Camp and backtrack

First entry to NAV, if no camp is stored, the current fix becomes CAMP. Camp persists in localStorage and is not silently moved the next time NAV opens. TRICORDER CAMP or the CAMP key marks home at the current fix. BACKTRACK (voice or key) points DEST at CAMP so bearing and the off-track corridor run toward home.

Walk out with no dest if you only need a breadcrumb; press BACKTRACK when you want the return line.

3.4 Setting a destination

Type or speak a dest, then GO. Accepted forms:

41.50N 74.01W

N 41.50 W 74.01

41.50, -74.01

87G8xxxx+xx (Open Location Code / plus code)

///word.word.word (what3words – needs the settings key)

Bare 41.50, 74.01 is east longitude (Central Asia), not Newburgh. Always send W or a minus on western longitudes. Hands-free speech often drops the decimal point (41 50 N 74 01 W); 0.6.41 rebuilds that as 41.50 / -74.01. Voice: TRICORDER NAVIGATE TO 41.50N 74.01W.

3.5 Coordinate matrix (0.6.44)

The CRT bezel is 230 px high and used to clip anything drawn under the map. CAMP / YOU / DEST lat-lon and the SAID echo now sit under STANDBY at the top of the tube:

LAT LON

CAMP 41.50xxx N 74.01xxx W

YOU ...

DEST ...

SAID 41.50N 74.01W

If DEST does not read 41.50000 N / 74.01000 W after a Newburgh command, the parse failed — read SAID before you walk.

3.6 Scrollable CRT (0.6.45)

In NAV the phosphor pane scrolls as one window: matrix, artificial horizon, north-up map, then GO / MARK / CAMP / BACKTRACK. Swipe on the tube, not on the keypad. NAV also raises the pane to 300 px so more of the map is in frame before the swipe.

3.7 Build notes 0.6.42–0.6.43

0.6.42 declared const strip twice inside paintNav and took the whole face down (one script block runs the instrument). 0.6.43 removed the duplicate. Do not sideload 0.6.42.

4. Face map

I Inquiry — Gemma on :8080. Wake words COMPUTER, HELP, DIAGNOSTICS, CLOCK, DICTATION, SETTINGS, BACK END.

II Sensor — camera + SmolVLM + flora/fauna. Hold is a toggle. Continuous machine vision writes the CRT without Kokoro.

III Sentry — continuous default. Weapon / possible-weapon / motion / animal / rain watches. Motion does not gate the caption loop.

IV Audio — YAMNet in-process. Optional BirdNET. Context watch (NOAA, ADS-B, citizen) when Earth link exists.

T / XLATE — offline speech-to-speech. SOURCE / TARGET wheels. AUTO swaps direction after each utterance.

H Horizon / NAV — first press horizon, second press walking NAV, third press off.

S Sat/Stars — live satellites then star map. O Sol — space weather + orbits + HF/VHF/UHF, then earth forecast, then moon phase, then off.

L E V P — log review, officer note, video log, play log. IPL / ALL abort in-flight work. STT kills speech only.

ROBO four-state cycle: dark = Kokoro · green ROBO = Android robot voice · yellow [H/F R] = robot + hands-free top-level listen · red [MUTE] = silent observer (screen and log only). Next touch returns dark / Kokoro.

5. Hands-free TRICORDER menu (0.6.32, still current)

Yellow [H/F R]: say TRICORDER then the function. Leave a mode with the face button. Speech that is not a TRICORDER line is ignored. Audio, translator, clock, and dictation own the microphone while they are up. Aliases include “trick order” and “tric order.”

TRICORDER SENSOR · AUDIO · SENTRY · TRANSLATOR / TRANSLATE · HORIZON · NAV · SATELLITES · SOL · CLOCK · DICTATION · SETTINGS · BACK END · DIAGNOSTICS · SUMMARIZE READINGS · HELP · CAMP · BACKTRACK · NAVIGATE TO ... · MARK.

6. Detector CRT, readings, flights (0.6.33–0.6.37)

MUTE is the fourth ROBO state. Audio / Sensor / Sentry keep raw percents; scroll the CRT up for them. Under that list a fused TOP 3. Number one HIGHLY LIKELY is the bottom bar (green when it leads). Cattle / moo is its own bucket; cat no longer matches “Cattle, bovinæ.”

Readings: SUMMARIZE READINGS [category] [day|week|month|all]. Touch a bar to drill. Flights add By tail and By airframe. A flights bar accepts only kind=flight; playback is by log id so Silence/Speech rows cannot leak into the timeline.

Overhead card: OpenSky + adsbdb. Photo from Planespotters / airport-data / JetPhotos. Touch photo for airframe facts, FROM/TO for the city dossier, DISMISS AIRFRAME to release the pin, BACK TO AIRFRAME to return from a city.

7. Clock, moon, places

CLOCK opens the GNSS face. Analog or digital, 12- or 24-hour. Voice: analog, digital, twelve hour, twenty four hour, timer coffee five minutes, cancel timer coffee, cancel all, explain, terminate. With a GNSS lock the face uses Android GnssClock. Display 4 Hz; hardware epoch about 1 Hz interpolated. Field clock, not a time standard. Field-log timestamps stay ordinary milliseconds.

Moon is the third press of Sol (or voice MOON). Logs and the horizon carry city/state when Geocoder has it, plus an offline plus code. Optional what3words key in settings.

8. Dictation

Wake: dictation or TRICORDER DICTATION. NEW, EDIT nnnn, LOG nnnn (CONFIRMED), SUMMARIZE nnnn, MINDMAP keyword, SPEAK nnnn, KILL nnnn (CONFIRMED), TERMINATE. ZULU then SAVE or ABORT. Buffer: CURSOR MINUS / PLUS word, BACKSPACE, NEWLINE, CUT / CUT WORD, PASTE, HOME / END. PUNCTUATION PERIOD (also bare PERIOD / DOT).

9. Watch and IPL

Sentry captions every dwell. Sure weapon: klaxon and WEAPON DETECTED spoken three times. Possible weapon: chime. Audio priority: gunfire ×3 + klaxon; scream; siren ×3; air / rail / ship; vehicles; named animal; humans ×3; entertainment. Fireworks can look like gunfire. IPL aborts in-flight work. Text longer than 80 characters is chunked for Kokoro.

10. Knowledge banks

MEMORY ALPHA is Simple English Wikipedia, cached on device. MEDICAL uses English Wikipedia for body and photograph; WikiMed is a title catalog only. One-shot: “memory alpha Airbus A320” or “medical pink eye.” TERMINATE closes.

11. Backends and first boot

Termux four-pack: ~/bin/tricorder start. Ports 8080 / 8081 / 8188 / 5000. Image sliders: steps and size (256 / 384 / 512). See TERMUX-BACKEND-INSTALL.md in the source tree.

On-device packs do not need Termux: YAMNet ships in the APK; BirdNET, flora/fauna, and translator language packs fetch on first Earth link if missing. The unit warns if Termux by F-Droid is not installed.

12. Hardware note

The reference computer is a Galaxy Z Fold 3 (Snapdragon 888, 12 GB). Budget tablets in the Onn class (Helio G80 or Snapdragon 685, 4–6 GB) are not a host for this integration.

13. Document control

Rev 0.6.45. On-device copy: assets/www/manual.html. This PDF is the complete field narrative through walking NAV. Prior complete Word guide: Rev 0.6.37. Source tree tag: project-tricorder 0.6.45.